movement. The signals produced by the movement is processed by an open-source p2p software (also developed by the university) to produce music in real time. Boney M would be proud.

Low-cost hand

Many ideas are born out of dire need; one such need for a robotic hand was felt by Richard Van of South Africa who lost four of his fingers in an accident. He collaborated with a Seattle-based prop designer to design prosthetics that could work like real hands and fingers but soon found out that it was unaffordable ($10,000 for every finger). MakerBot came to the rescue and donated their 3D printers for the project. This brought down the cost of the prosthetic to around $150.

Pimp up your limb

Face it, you know you were thinking about it and knew it was coming. Enter prosthetics add-ons. Why hide it when you can flaunt it? San Francisco-based Bespoke Innovations is taking 3D printing to an artistic level by designing customisable fairings for prosthetic legs. Their fairings are specialised coverings that surround an existing prosthetic leg and are customised to reflect the personality of the user. The fairings can be enhanced with graphics, patterns, tattoos and even different materials like leather, nylon fabric and chrome plating (thats right, chrome!).

It's alive!

Not exactly a Frankensteinian venture but around the world, prosthetics are being printed with living cells (thats right, living cells!).

The first, Super Ear was created by Researchers from Princeton and Johns Hopkins. They came together to 3D print an ear which was seeded with cartilage cells. Now the reason why we called it a Super Ear is that they have also included a microwave-sensitive antenna into it. Theoretically, with a few more additions, the ear would be able to pick up electromagnetic signals like radio, TV, Wi-Fi, etc. Ironically, the ear cannot be used for hearing because it does not have a microphone since a microphone cannot be printed. But we'll soon have fully-functioning ears in the market, too. Researchers in China have created a 3D printer called Regenovo. Unlike other 3D printers that use plastic and other material to print, the Regenovo prints using living tissue. The researchers also state that fully-functioning 3D-printed organs could soon be a reality. In a world where people perish due to lack of organ donors, this would definitely solve most of the problem.

Prosthetic realism

All prosthetics suffer from one common issue - aesthetics. No matter how good they look, when it comes to the real thing, they don't match up. Solving a part of this problem is a consultancy in England called Fripp Design. They use sophisticated techniques to collect the 3D data to create the prosthetic organ and customise it to make it look real it by adding wrinkles, pores, colour, etc to it, depending on the age of the person for whom it is being made. For 3D printing the organ, they use a custom formulated starch substrate mixed with medical-grade silicone to give the organ flexibility and robustness. The results are quite astounding and look very realistic.

Mind-Controlled Prosthetics

What good is a prosthetic if you can't make it move to your will? There are several researches that are working just to solve this problem. No more walking with a stiff joint. Long John Silver would have loved to hop around on one of these. There are innumerable researches going on around the world when it comes to mind-controlled prosthetics:

Nerve/Muscle Interface

Both military funded operations, the US has been developing bionic arms and legs

Prosthetic limbs advance further by the day

Mind-control on Prosthetics like a natural limb?

which use the nerve signals in the limbs to move the prosthetic instead of tapping into the brain for the signals. The first is DARPA's Reliable Neural-Interface Technology (RE-NET) programme which uses the signals that are already coming through the nerves and muscles to move the prosthetic arm. The second is the US army-funded research into a mind controlled bionic limb that enables a person to walk, climb stairs, wiggle the feet, etc. With just 1.8 per cent chance of error, the bionic leg dances to the wearer's tunes and moves when the wearer thinks it. The bionic leg is interfaced to the leg through nerves rewired to the hamstring and linked through sensors. Currently, this project has seen investments to the tune of $8 million.

I can feel again

This is something that every amputee wants, to feel the lost limb again. While it may not be possible currently to regrow a limb, researchers are directly tapping into the brain to send the sensation of touch to it from the limb.

Although the research is currently being conducted on rhesus macaques, it won't be long before the results are applied to developing advanced prosthetics that allow one to feel the sensation of touch. The research was carried out by inserting electrodes into the macaques' brains into regions that mimic the sensation of touch and also reflect the location of the sensation trigger. Through the electrodes they simulated the feeling of being touched and rewarded the macaques with treats for

Photo courtesy: Raytheon Company